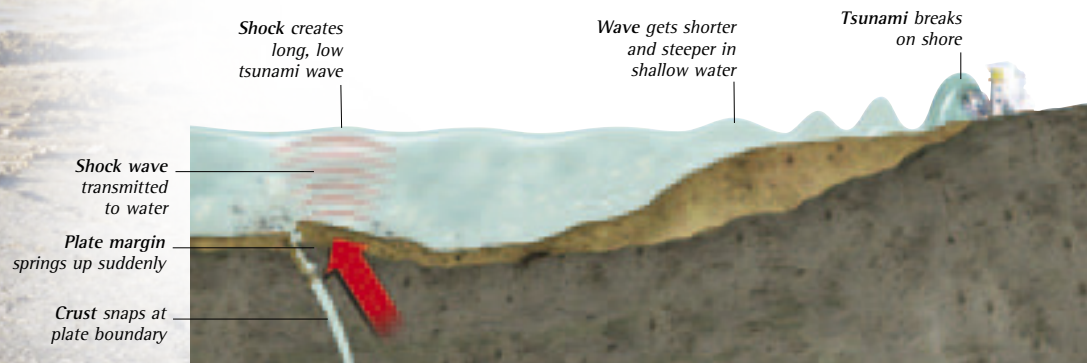


TSUNAMIS

When an earthquake occurs on the ocean floor, it creates fast-moving shock waves called tsunamis. On the open ocean the waves are broad and low, with a very long wavelength from crest to crest. When they reach shallower water the waves get shorter and steeper, piling up into very high crests and equally deep troughs. Eventually, a trough reaches the coast, causing the sea to draw back by many yards. Then the wave crest strikes, often with devastating power.

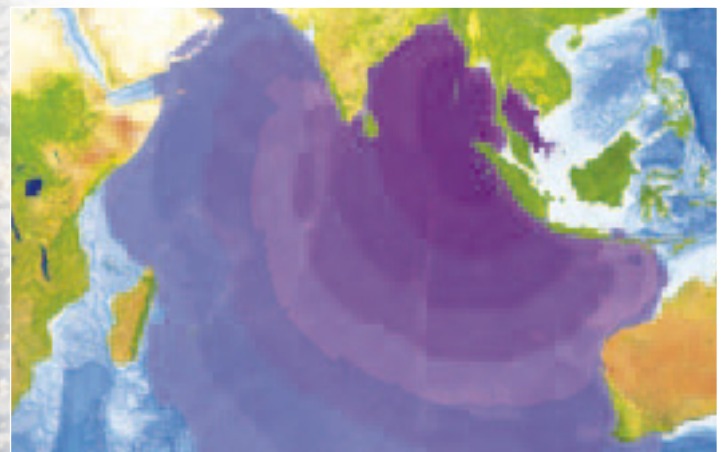


▲ SHOCK WAVE

Many earthquakes are caused by sections of the Earth's crust slipping suddenly along plate margins. In subduction zones, the steady movement of the sinking plate can drag the edge of the other plate down and then suddenly release it. The plate springs up, carrying the water up with it, which then flows out sideways as a tsunami.

◀ STRAIN AND SNAP

The plates of the Earth's crust are always moving. If they slip steadily, the movement causes small, regular tremors. But if the plates become locked against each other, the strain builds until something snaps. If the crust has distorted by 10 ft (3 m) by the time this happens, the rocks move this distance in a few seconds to release the tension. The colossal shock of this movement causes an earthquake.



▲ RADIATING WAVES

The 2004 Asian tsunami was caused by an earthquake in the Java Trench off northern Sumatra. The rupture was 750 miles (1,200 km) long, and the rocks moved vertically by 50 ft (15 m). The resulting tsunami radiated outward across the Indian Ocean at 500 miles (800 km) per hour. Each band shows the distance traveled in one hour.